SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C01_AT2 — DATA INTERPRETATION EXERCISE

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO1 –Apply (BL3)	**Assessment:**	Assessment Tool 2 – Data Interpretation
Weightage:	20% of CIA	**Total Marks:**	20(2 Questions × 20 Marks)
CO1 Statement:	*Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)*		
Student Name:	Tejaswini P	**Reg. No.:**	192324046
Section / Batch:	2023 / B.tech AIDS	**Date:**	28/7/2026

Code of Conduct

I Tejaswini P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tejaswini

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.

Annexure A – DATA INTERPRETATION

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Data Interpretation Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT2 Data Interpretation Exercise problems, in which students interpret a described data visualization (data types, aesthetic mappings, scales, coordinate systems, and color scales) and answer accompanying sub-questions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough, accurate understanding of data types (nominal, ordinal, numerical), aesthetic mappings, scales, coordinate systems, and color scales as they apply to the given visualization scenario.	Good understanding of the relevant visualization concepts, with only minor gaps in identifying data types, aesthetic mappings, or scale choices.	Basic understanding of visualization concepts, but with some misconceptions about data types, aesthetics, coordinate systems, or color scales.	Poor understanding of core visualization concepts; major misconceptions about data types, aesthetics, coordinate systems, or color scales.
Explanation	**25%**	Provides detailed, well-reasoned explanations for each sub-question, using specific details from the scenario and correct visualization terminology.	Provides clear explanations with adequate justification, referencing the scenario appropriately for most sub-questions.	Explanations are limited or superficial; lacks depth or fails to consistently connect reasoning to the given scenario.	Explanations are poor, unclear, or missing; answers are unsupported assertions with no reasoning shown.
Application	**20%**	Effectively applies visualization concepts to interpret the specific scenario, drawing accurate, well-reasoned conclusions from the described chart or data pattern.	Applies concepts to interpret the scenario correctly, with only minor errors in reasoning or in the conclusions drawn.	Limited application of concepts to the scenario; conclusions are weakly supported or only partially correct.	Fails to apply concepts to the scenario, or the conclusions drawn are incorrect or irrelevant to the data shown.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Organization	**15%**	Answers are well-structured, addressing each sub-question (a, b, c) in a clear, logical sequence with coherent flow throughout.	Answers are organized and address each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but answers lack clarity or a logical sequence.	Poorly organized; answers are incoherent, and sub-questions are left unaddressed or answered out of order.
Accuracy	**10%**	All parts of the answer — data type identification, aesthetic/scale justification, and final interpretation — are completely accurate.	Mostly accurate, with only minor omissions or a small error in one part of the answer.	Partially correct; some key points (e.g., data type, aesthetic justification, or interpretation) are missing or incorrect.	Answers are largely incorrect, with major gaps in identifying data types, aesthetics, scales, or drawing conclusions.

Scoring Guide

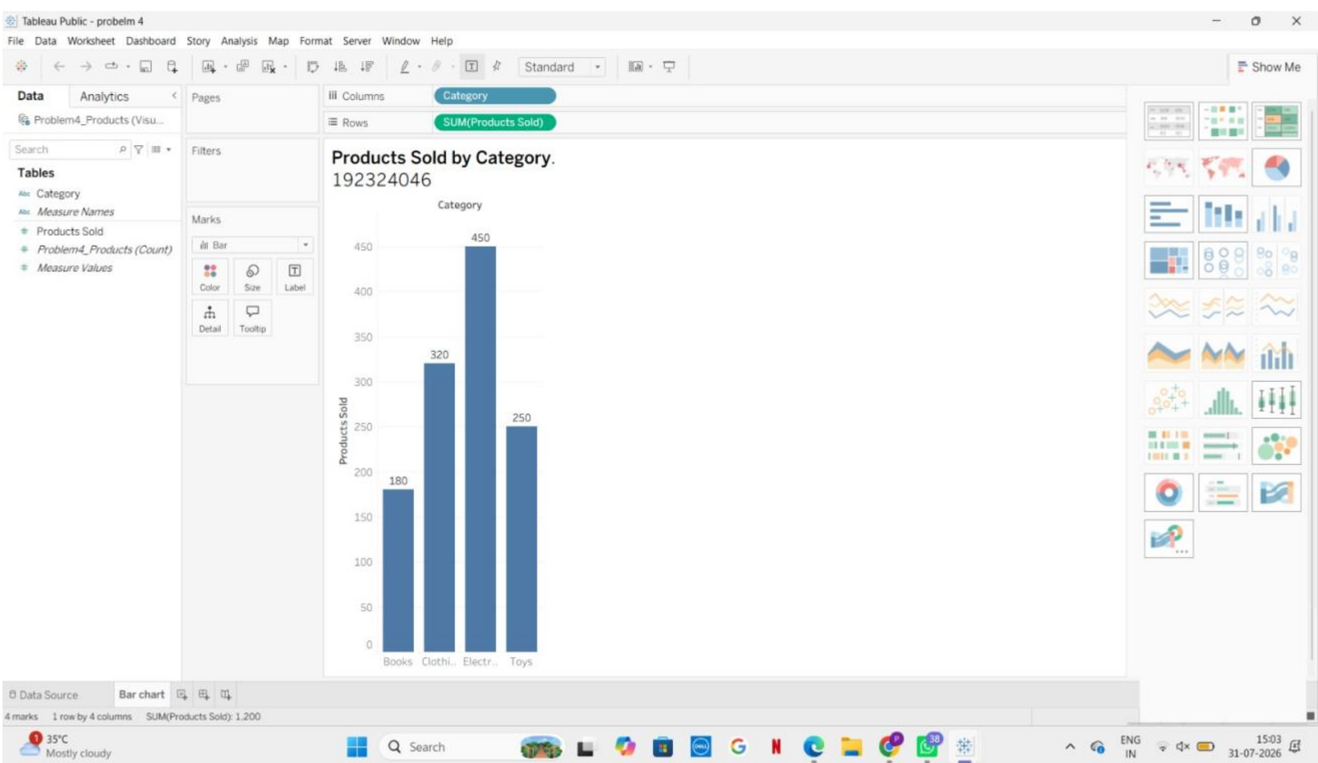
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Problem 4 (10 Marks)

Scenario: A bar chart shows the number of products sold in four categories (Electronics, Clothing, Books, Toys), with bar height mapped to unit count and bar width kept constant.

a) Which aesthetic mapping encodes the primary data value in this chart?
The bar height (length) encodes the primary data value, which is the number of products sold (unit count). The x-axis represents the product categories (Electronics, Clothing, Books, Toys), while the y-axis shows the number of units sold.
b) Would size (bar width) have been a valid additional aesthetic to encode a second variable, such as profit margin? Explain.
Yes, it is possible, but it is generally not recommended. Changing the bar width to represent profit margin could confuse viewers because bar charts are designed for comparison using bar height. Different bar widths make the chart harder to interpret and may distract from the primary comparison. A better choice is to use color, labels, or a separate chart to represent profit margin.
c) If "Electronics" has the tallest bar but the smallest profit margin (not shown), what interpretation risk does this chart create for a viewer?
The chart may lead viewers to incorrectly assume that Electronics is the best-performing category overall because it has the highest sales. Since the profit margin is not shown, viewers cannot see that Electronics actually earns the lowest profit per sale, resulting in a misleading interpretation of the category's overall performance.



Problem 14 (10 Marks)

Scenario: A GPS tracking app plots a delivery driver's location every 10 minutes over a single day using standard Cartesian (X, Y) coordinates on a city map background.

a) What do the X and Y axes represent in this context?

- X-axis: Horizontal position (east-west location or longitude) of the driver on the city map.
- Y-axis: Vertical position (north-south location or latitude) of the driver on the city map.

b) If the driver returns to the same starting point at the end of the day, how would that be reflected in the coordinate data?

The first and last coordinate pairs would be the same (or nearly the same due to GPS accuracy). For example, if the trip starts at (10, 20), it would also end at (10, 20), showing that the driver returned to the starting location.

c) Would polar coordinates be a more or less natural choice for this specific scenario? Justify your answer.

Polar coordinates would be less natural. GPS locations on city maps are best represented using Cartesian (X, Y) coordinates because they directly show positions on a map. Polar coordinates (distance and angle from a fixed point) are more suitable for situations involving circular or radial movement, not for tracking routes through city streets.